

CATTAIL COMFORT

Natural Insulation Redefined

Developed in collaboration with Timelab and Smissenbroek—a material workspace & regenerative farming initiative, respectively—this project explores the potential of Narrowleaf Cattail (*Typha angustifolia*) as a sustainable insulation material. The project originated from a practical challenge: improving thermal comfort in the poorly insulated home of Smissenbroek founder Isabelle d'Ursel.

Using cattail fluff's natural hydrophobic and insulating properties, I developed biodegradable textile wall panels that reduce heat loss through radiation. The design was informed by historical research into textile wall coverings, which were traditionally used to improve thermal comfort by retaining radiant heat rather than simply heating the air. This approach offers a more energy-efficient and human-centered alternative to conventional heating systems.